propulsion. Its first commercial products will be battery-powered aircraft intended for shorter-range operations. In parallel, the startup is developing hydrogen fuel-cell systems for future long-range VTOL platforms targeting a range of approximately 250km. "The hydrogen architecture follows a parallel hybrid configuration in which both the battery and fuel cell supply power during take-off and landing, while the fuel cell alone powers cruise flight. This is intended to combine the high power required for vertical lift-off with the higher energy density and endurance benefits of hydrogen during forward flight," explains Pranay.

The startup has already built and ground-tested a hydrogen fuel-cell powerplant and high-pressure carbon-fibre hydrogen tanks.

BluJ currently employs about 50 full-time staff and is working closely with regulators on approvals for beyond-visual-line-of-sight and long-range cargo operations. "Regulatory engagement is particularly important because commercial VTOL operations require approvals covering airworthiness, operational procedures, remote operations, and safety management," says Pranay.

The startup's immediate plans include establishing an operational base for cargo services, expanding the Reach product line, and advancing its 250km hydrogen-electric aircraft programme. It also intends to establish a larger dedicated production facility as manufacturing volumes increase. In the longer term, once its cargo operations have gained sufficient maturity and public confidence, BluJ plans to enter the passenger mobility market.

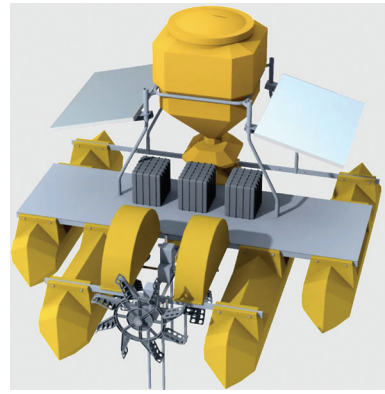
Explaining the startup's name, Pranay says 'BluJ' is inspired by the Blue Jay bird associated with Telangana, reflecting the founders' local roots and the company's origin in the state.

—Nitisha Dubey

2 AQUAEXCHANGE BRINGS MONITORING AND AUTOMATION TO SHRIMP FARMING

The startup has developed an AI- and IoT-enabled platform that helps shrimp farmers manage farms, access financing, monitor operations, and connect with markets.

The startup was founded by engineers Pavan Kosaraju, Hemasundar D, Kareemullah Md, and Kiran Kumar, who grew up in small towns and villages along the coastal belt of Andhra Pradesh, where shrimp farming has been part of daily life for more than 30 years. Their goal was to create a platform that connects everyone involved in shrimp farming—from farmers and hatcheries to exporters, banks, insurers, and retailers. That idea also inspired the name AquaExchange.



AquaBot, a solar-powered automatic feeder



PowerMon, an IoT-based power and aeration management system

AquaExchange provides a full-stack technology platform for the shrimp aquaculture ecosystem. The company offers solutions in three main areas. The first is farm management, where it helps farmers improve day-to-day operations, increase efficiency, reduce mortality, and use predictive tools to improve farm management.

The second area is finance and risk management. AquaExchange enables banks to provide direct loans to farmers through its platform. AquaExchange built a data-driven platform that gave banks the structured information they needed while helping farmers access loan products suited to their needs.

The third area is market connectivity. Since more than 90% of India's farmed shrimp is exported, the platform connects farmers with processors, exporters, and global buyers.

Speaking about the product line, Pavan explains, "Shrimp farming needs continuous monitoring and automation, so hardware is a core part of what we do. We have developed a portfolio of around 17 IoT products, all designed, developed, and manufactured at our facility near Vijayawada. We also own the intellectual property for all these products."

One of its key products monitors aerators, which supply oxygen to shrimp ponds. Since dissolved oxygen levels are critical for shrimp survival, even a delay of two to